
Freight Rate Prediction Machine Learning Assessment Report

Time-aware regression for freight load pricing

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Model:	LightGBM Regressor
Local validation:	Chronological holdout (Sep–Oct 2025)
Final prediction set:	12,000 loads
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Executive summary. A deployment-oriented LightGBM regression pipeline was developed to predict freight rates from shipment, route, equipment, weight, geographic, and temporal information. A chronological validation design was used to better approximate future-period inference. The local holdout achieved an MAE of \$116.97, RMSE of \$636.57, and MAPE of 5.10%. The final model was retrained on all 48,000 labeled observations and used to produce 12,000 validation predictions plus the required December 2025 fixed-route forecast.

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1 Problem Statement

The objective of the assessment is to predict the posted freight rate for each load in the supplied validation data and to generate a daily predicted-rate series for the fixed December 2025 scenario. The modeling workflow was designed around three priorities:

- preserve as much labeled information as possible rather than unnecessarily sampling the data;
- mimic the temporal nature of the hidden evaluation through a chronological holdout; and
- keep preprocessing consistent between training, validation, and final inference.

The final output file contains one positive `predicted_rate` for each unique `load_id` in the 12,000-row validation set.

2 Data Preparation and Quality Handling

2.1 Weight cleaning

The raw weight feature contained missing values and invalid negative values. Negative weights were treated as sign errors and transformed using absolute value. Missing weights were imputed using the median cleaned weight for the corresponding equipment type, calculated only from the training portion when evaluating the chronological holdout. If an equipment-specific median was unavailable, the global training median was used.

2.2 Identifier handling

The `load_id` field was excluded from model features because it serves as an identifier rather than a predictive measurement.

2.3 Deployment-consistent feature selection

The fields `market_index` and `quote_signal` were not used in the main deployment model because they are not available in the December prediction input. This avoids maintaining a separate feature pipeline that cannot be applied consistently to the required December scenario.

2.4 Temporal feature engineering

The shipment date was converted into day-of-week and periodic calendar representations. Month and day-of-year were represented using sine/cosine pairs so that the model can learn cyclic calendar patterns without treating December and January as maximally distant values.

2.5 Geographic feature engineering

Latitude and longitude were retained for both pickup and delivery locations. Two additional directional differences, `lat_diff` and `lon_diff`, were created. For the December route, city coordinates were recovered from the development-data city-coordinate lookup.

3 Feature Set

The final model uses 14 features:

Feature group	Variables
Shipment	distance, weight_clean, equipment
Geography	pickup_lat, pickup_lon, delivery_lat, delivery_lon
Route deltas	lat_diff, lon_diff
Calendar	month_sin, month_cos, doy_sin, doy_cos, dow

Table 1: Feature groups used by the LightGBM model.

4 Validation Strategy

A random train/validation split would mix earlier and later dates and could produce an overly optimistic estimate when the actual task is to predict future loads. Therefore, a chronological split was used.

Partition	Rows	Share	Date range
Training	38,477	80.2%	Jan 1 – Aug 31, 2025
Local validation	9,523	19.8%	Sep 1 – Oct 31, 2025
Total	48,000	100.0%	Jan 1 – Oct 31, 2025

Table 2: Chronological development split used for local model validation.

All preprocessing statistics needed by the holdout model, including weight medians, were fitted on the training portion before being applied to the September–October holdout.

5 Model

The selected estimator is a LightGBM gradient-boosted decision-tree regressor. It is well suited to heterogeneous tabular data, nonlinear interactions, and categorical features while remaining fast enough for iterative assessment work.

Hyperparameter	Value
Objective	regression_l1
Number of estimators	600
Learning rate	0.03
Number of leaves	31
L2 regularization (reg_lambda)	10.0
Random seed	42
Categorical feature	equipment

Table 3: Final LightGBM configuration.

The L1 objective aligns directly with absolute-error minimization and is less dominated by extreme residuals than a squared-error objective.

6 Local Validation Results

Mean Absolute Error
\$116.97

Root Mean Squared Error
\$636.57

Mean Absolute Percentage
Error
5.10%

The MAE indicates that the typical absolute deviation on the September–October holdout was approximately \$117. The higher RMSE relative to MAE indicates that a smaller set of errors are substantially larger than the typical error. The MAPE of 5.10% provides a scale-normalized view of holdout performance.

These are **local chronological holdout metrics only**. They are not the final Spotter evaluation metrics; the supplied scorer explicitly states that final validation metrics are calculated after submission.

7 Feature Importance

The fitted final LightGBM model reported the following split-based feature importances. These values are useful for model inspection, although they should not be interpreted as causal effects.

Feature	Importance	Feature	Importance
distance	4030	pickup_lat	1066
weight_clean	1908	delivery_lat	1059
doy_cos	1778	lon_diff	757
doy_sin	1343	lat_diff	745
equipment	1333	month_cos	715
pickup_lon	1158	dow	675
delivery_lon	1150	month_sin	283

Table 4: LightGBM feature importance from the final model trained on all labeled observations.

Distance had the largest split-based importance, followed by cleaned shipment weight and annual cyclic date features.

8 Final Training and Validation Predictions

After the modeling choices were fixed, preprocessing state and the LightGBM model were refitted using all 48,000 labeled rows. The final model then generated predictions for all 12,000 validation loads.

The produced submission file passed the following internal checks:

- exactly 12,000 rows;
- exactly two columns in the required order: `load_id`, `predicted_rate`;
- 12,000 unique load identifiers;
- no missing prediction values; and
- all predicted rates strictly positive.

Validation prediction statistic	Predicted rate (\$)
Mean	2,247.91
Standard deviation	1,297.35
Minimum	211.06
25th percentile	1,214.68
Median	1,928.30
75th percentile	3,202.07
Maximum	6,527.13

Table 5: Summary of the 12,000 submitted validation predictions.

9 December 2025 Fixed-Route Forecast

The assessment requires one prediction for every date from December 1 through December 31, 2025 under the following fixed conditions:

Input	Fixed value
Pickup	Lexington
Delivery	Fort Wayne
Distance	360 miles
Equipment	Dry Van
Weight	32,000 lb
Date	December 1–31, 2025

Table 6: Fixed December prediction scenario.

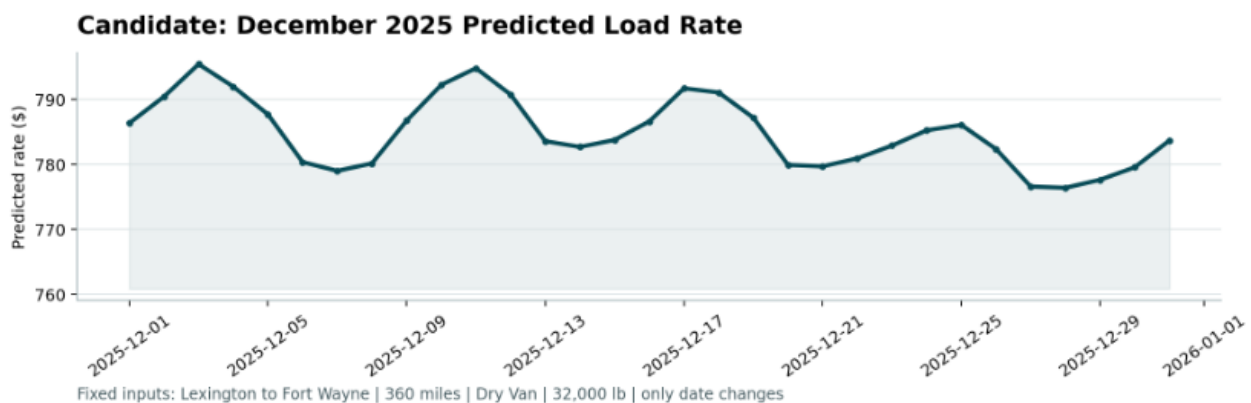


Figure 1: December 2025 predicted load rate generated from the completed fixed-route predictions.

The forecast shows a moderate day-to-day seasonal pattern around the high-\$700 range for the fixed route. Because all operational inputs are held constant and only the date changes, the variation in Figure 1 is attributable to the model’s temporal features.

10 Reproducibility and Output Validation

The notebook pins the core numerical and plotting libraries to the assessment-compatible ranges and additionally uses LightGBM and scikit-learn for model training and evaluation. A fixed random seed

of 42 is used for reproducibility.

The supplied scorer is used after prediction generation to validate both the 12,000-row submission and the 31-row December file. It also produces the required `candidate_december.png` visualization.

11 Limitations and Possible Improvements

- **Single temporal holdout.** The September–October holdout is deployment-relevant, but additional rolling or expanding-window validation would provide a more robust estimate of temporal generalization.
- **Large-error sensitivity.** The gap between MAE and RMSE indicates that a small number of observations have much larger residuals. Targeted analysis of those routes and rate regimes could identify data-quality issues or missing explanatory variables.
- **Unavailable deployment features.** `market_index` and `quote_signal` were intentionally excluded for deployment consistency. If equivalent forward-looking values became available at prediction time, they could be evaluated in a leakage-safe pipeline.
- **Route representation.** Future iterations could compare coordinate features with route-level historical aggregates, provided those aggregates are constructed strictly from past observations.

12 Conclusion

The final solution uses a leakage-aware, time-based validation design and a compact LightGBM model trained on all available labeled observations after model selection. The local chronological holdout achieved MAE \$116.97, RMSE \$636.57, and MAPE 5.10%. The final pipeline generated all 12,000 required validation predictions and the complete 31-day December fixed-route forecast in the formats expected by the assessment scorer.

Files accompanying this report:

- `main.tex` – LaTeX report source;
- `figures/candidate_december.png` – required December chart;
- `data/validation_predictions.csv` – final validation predictions;
- `README_COMPILE.txt` – compilation instructions.